

MOHSEN FEKRI

AI & DATA | MACHINE LEARNING | COMPUTER VISION

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PROFESSIONAL SUMMARY

AI and machine learning practitioner with hands-on experience in faculty-led research, spatio-temporal medical-image segmentation, credit-risk modelling, and enterprise RAG. Pursuing a BAsC in Computer Science for Software Development at York University, Lassonde School of Engineering, with a 6.7/9.0 GPA. Comfortable discussing model evaluation on imbalanced data, model drift in production, and inference optimization. Familiar with EY.ai, EYQ, FlexiGenAI, and Responsible AI, with interests aligned to EY AI & Data, Digital Engineering, and Technology Risk.

TECHNICAL SKILLS

Programming & Data: Python, SQL, Java, JavaScript, Pandas (DataFrames), NumPy, Scikit-learn, XGBoost

AI & Machine Learning: PyTorch, MONAI, computer vision, classification, segmentation, spatio-temporal modelling, model evaluation (precision, recall, F1, ROC-AUC), imbalanced data, model drift, explainable AI

GenAI & Development: RAG, MCP (Model Context Protocol), OpenAI-compatible API standard, vector databases, LangChain/LlamaIndex, FastAPI, Streamlit, Gradio, Docker, Kubernetes, TensorRT, Git, Responsible AI

AI RESEARCH EXPERIENCE

Teaching Assistant & AI Research Contributor | York University | 2025

Supervisor: Prof. Navid Mohaghegh | UGRA: Spatio-Temporal Ultrasound Needle Segmentation

- Teaching assistant and research contributor on AI-assisted needle localization in ultrasound sequences.
- Contributed to a PyTorch and MONAI segmentation pipeline combining Swin Transformer **spatial** feature extraction with ConvLSTM **temporal** modelling across frames.
- Worked with sequential ultrasound inputs and supported preprocessing, model development, and frame-level segmentation evaluation using Dice and IoU.

SELECTED AI PROJECTS

AI Credit Risk & Loan Decision Support System | Python | Scikit-learn | XGBoost | Explainable AI | 2026

- Built a classification pipeline that analyzes borrower credit history and financial attributes to estimate loan eligibility and default risk.
- Handled class imbalance in the default population and evaluated performance across precision, recall, F1-score, ROC-AUC, and confusion-matrix analysis, weighing false positives against false negatives.
- Surfaced influential features for transparent review and presented predictions as decision support rather than automatic approval, keeping final lending decisions with a human reviewer.

Enterprise Document Question-Answering Assistant | Python | RAG | MCP | FastAPI | Gradio | RAGAS | 2026

- Built a RAG application that ingests PDF and Word files such as company policies, audit procedures, risk reports, and cybersecurity documentation.
- Exposed retrieval as **MCP** tools behind an OpenAI-compatible API so the assistant can be driven by agentic clients.
- Retrieved supporting passages, returned page-level citations, and abstained when the document collection did not contain enough evidence for an answer.
- Compared chunk sizes and retrieval methods using RAGAS faithfulness and retrieval-quality evaluation, with role-based permissions.

Medical Image Cancer Detection Prototype | Python | PyTorch | EfficientNet-B0 | TensorRT | Streamlit | 2025

- Built a CNN-based prototype that standardizes each medical image to 224 x 224 RGB, produces a sigmoid probability, and applies an adjustable classification threshold.
- Optimized inference with **TensorRT** quantization and pruning to reduce model size and latency while tracking the accuracy trade-off.
- Added a Streamlit interface and visual interpretability for reviewer oversight, with output clearly identified as educational and research-only rather than diagnostic.

EDUCATION

York University, Lassonde School of Engineering | Toronto (Markham Campus) | **Expected 2029**

Bachelor of Applied Science (BAsC), Computer Science for Software Development, Co-op | GPA: 6.7/9.0

ADDITIONAL EXPERIENCE

Operations Assistant | York University | **May 2026 - Sep 2026**

Program & Administrative Assistant | YMCA of Greater Toronto | **Dec 2024 - Jan 2026**

IT Support & Lab Assistant, Volunteer | Toronto Public Library | **Mar 2023 - Sep 2023**

CERTIFICATION IN PROGRESS

AI-103: Developing AI Apps and Agents on Azure | Microsoft | **In Progress**